SUMMARY · MK 5 DECISION MAKING & NEGOTIATION · WEEK 1

Introduction to Decision Making: Heuristics & Biases

Why study decision making: people use heuristics, fall into biases and have bounded rationality. Class games, intuition, the three Tversky–Kahneman heuristics, Gigerenzer on risk vs uncertainty, definitions, decision levels, and descriptive/normative/prescriptive approaches.

Material

1 What the class games show

Games show two thinking systems, anchoring (both products = 40,320), persistent illusions, intuitive errors (5 machines → 5 minutes), framing (both Hitler dates false) and priming (S__P → SOUP).

2 Two thinking systems at a glance

Deep dive

System 1 — fast

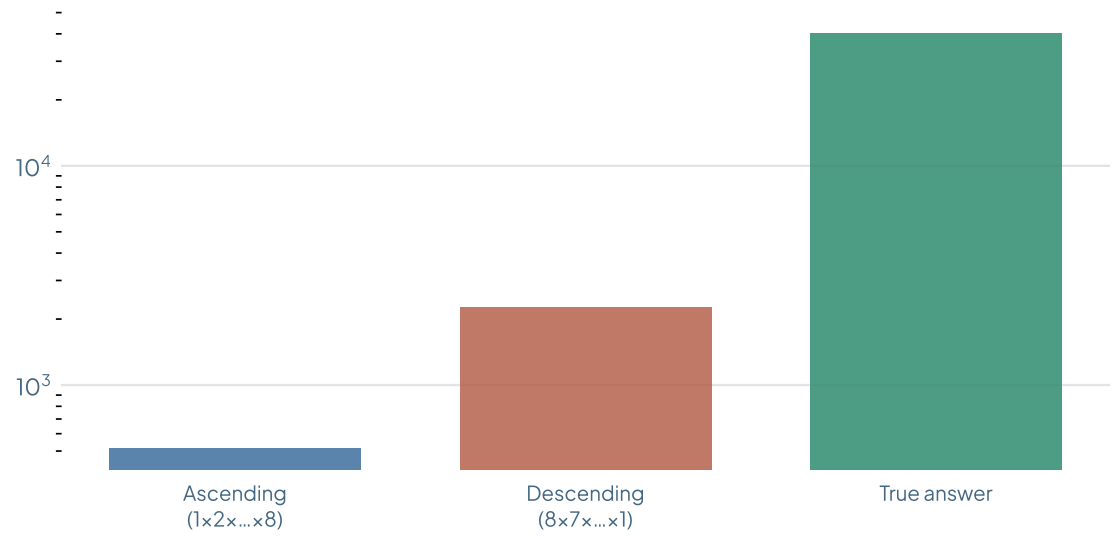
Automatic, effortless, parallel, emotional.

vs

System 2 — slow

Deliberate, effortful, serial, limited.

Anchoring: median guesses vs the true answer



Tversky & Kahneman (1974). Log scale.

Median guesses in Tversky & Kahneman (1974): 2,250 (descending) vs 512 (ascending); the true answer is 40,320. Adjustment from an anchor is insufficient.

3 Intuition and heuristics

Intuition is subconscious, parallel, emotional pattern matching; useful under time pressure, change, poor structure, ambiguity or no precedent. Heuristics: availability, representativeness, anchoring & adjustment — useful shortcuts that bias judgement when careful statistical reasoning is needed. Gigerenzer: under risk use models; under uncertainty robust heuristics often win.

4 Heuristic → bias → antidote Deep dive

Heuristic	Bias	Antidote
Availability	Over-weighting vivid events	Look at base rates
Representativeness	Base-rate and sample-size neglect	Ask about proportions and n

Anchoring	Insufficient adjustment	Estimate first; use a counter-anchor

Linda

"Bank teller and feminist" cannot be more likely than "bank teller" — the conjunction fallacy.

↕ Top row: stable world · bottom: fast-changing

Risk
Statistics and optimisation.

Uncertainty (stable)
Simple heuristics work.

Risk (changing)
Recalibrate models.

Uncertainty
Fast-and-frugal heuristics can win.

↔ Left column: probabilities known · right: unknown

5 What decision making is, and its levels

Definitions from Simon, Tryfos, Wang & Ruhe. Strategic decisions are broad, complex, enduring, delayed, contested and political. Descriptive (what people do), normative (what they should do in theory), prescriptive (what they should and can do — the course focus). Rational decision making weighs criteria and picks the highest utility.

6 Decision levels and approaches to studying decisions

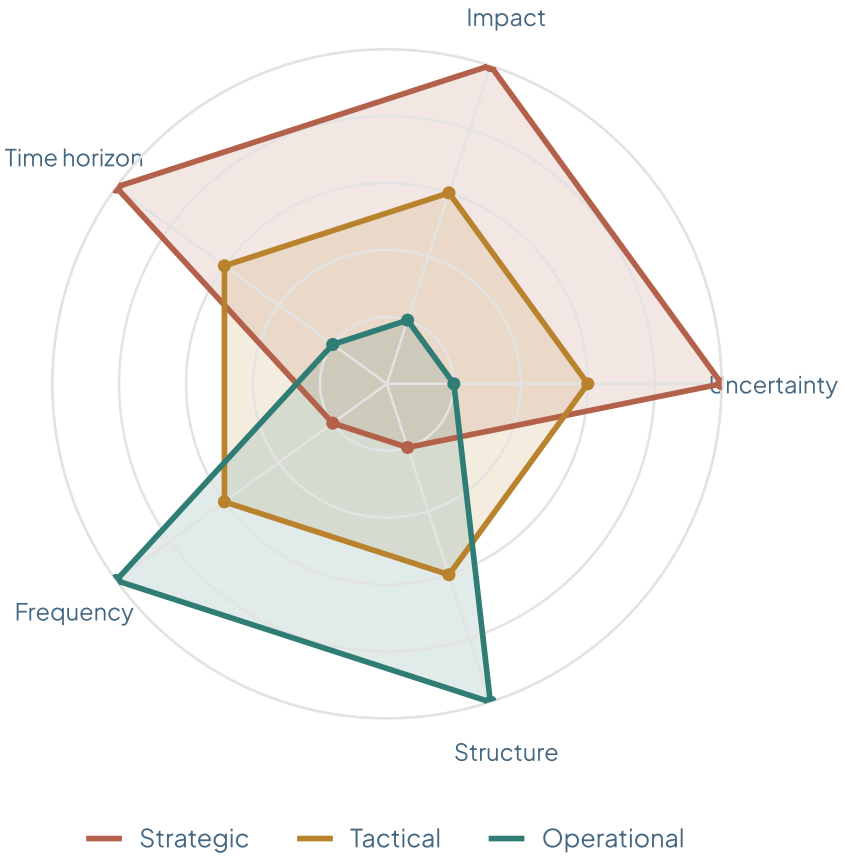
Deep dive

Strategic — top management, long-term, unstructured

Tactical — middle managers

Operational — supervisors, routine

Decision traits by level (conceptual, 1–5)



Descriptive
How people decide.

Normative
How rational agents should decide.

Prescriptive
Practical tools to help real people decide better.

7 Other biases that often appear in managerial decisions

Besides the three Tversky–Kahneman heuristics, several biases appear very often in organisations. All come from System 1 working fast without System 2 checking it.

Bias	What happens	Office example	Remedy

Framing	Choices change only because of how they are presented (gain vs loss)	"The programme saves 200 of 600 people" is preferred to "400 of 600 people die" — though they are the same	Rewrite the choice in the opposite frame and check whether your choice holds
Loss aversion (prospect theory)	The pain of a loss is about twice as strong as the pleasure of an equal gain	Rejecting a positive-EMV project because there is a small chance of loss	See the decision as part of a portfolio of many decisions, not one bet
Overconfidence	Too sure of one's own estimates; ranges too narrow	Estimating a project "done in 6 months" when similar projects average 10	Outside view (data on similar projects), ask for a 90% range
Confirmation bias	Only seeking evidence that supports the first opinion	A manager reads only the reports that favour a preferred vendor	Consider the opposite; appoint a devil's advocate
Sunk cost & escalation of commitment	Keep investing because "so much has already been spent"	An IT project that keeps getting funded although it is clearly failing	Ask: "If I were starting from zero today, would I invest?"
Hindsight bias	"I knew it all along" once the outcome is known	Blaming a decision that was reasonable when it was made	Record the reasons and information at decision time (a decision journal)
Status quo bias	Preferring no change	Keeping an old system although an alternative is clearly better	Treat the status quo as one more alternative that must also be evaluated

Why this matters for the coming weeks

The prescriptive tools (KT, VFT, SMART, decision trees, AHP) are basically ways of forcing System 2 to work: writing goals, alternatives, probabilities and weights explicitly so biases become easier to see and discuss.

Key takeaways

1. Bounded rationality, heuristics, biases.
2. Availability, representativeness, anchoring.
3. Risk ≠ uncertainty.
4. Descriptive, normative, prescriptive.
5. Strategic decisions are broad and irreversible.

Case studies

Case & worked answers

Reflection: easiest vs hardest decision

Easy decisions are repetitive, low-stakes, reversible; hard ones are one-off, multi-stakeholder, uncertain and irreversible — the same traits as strategic decisions, which call for prescriptive tools. Ask which heuristic shaped your own hard decision.

Practice questions

Try answering first. The answer key is below each question.

1. Judging plane-crash risk from news coverage is…

A. Anchoring
B. Availability
C. Representativeness
D. Satisficing

Answer: B. Easy to recall ≠ frequent.

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2. 5 machines, 5 min, 5 widgets → 100 machines, 100 widgets takes…

A. 100 min
B. 20 min
C. 5 min
D. 1 min

Answer: C. Each machine makes one widget per 5 minutes.

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3. The Linda problem shows which heuristic?

A. Availability
B. Representativeness
C. Anchoring
D. Affect

Answer: B. Similarity beats probability logic.

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4. Per Gigerenzer, simple heuristics shine under…

A. Risk
B. Uncertainty
C. Certainty
D. Never

Answer: B. Less is more under uncertainty.

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5. The approach offering practical tools for real people is…

A. Descriptive
B. Normative
C. Prescriptive
D. Intuitive

Answer: C. Prescriptive bridges the two.

Sources

- [Material] Week 1 - Intro to Decision Making (Dr. Valid Hasyimi)
- Bazerman, Judgment in Managerial Decision Making, Bab 1–2
- Gigerenzer, Simple Heuristics That Make Us Smart